RESPONSIBLE AI

Explainable AI Visual Guide

Opening the black box:
SHAP, LIME, saliency maps — and their limits.

ANIMATED WALKTHROUGH
Step through pages to play the animation

STEP 1/5

Global vs Local Explanations

Two questions, two tool families.

Global — 'what does the model rely on overall?' — feature importance.

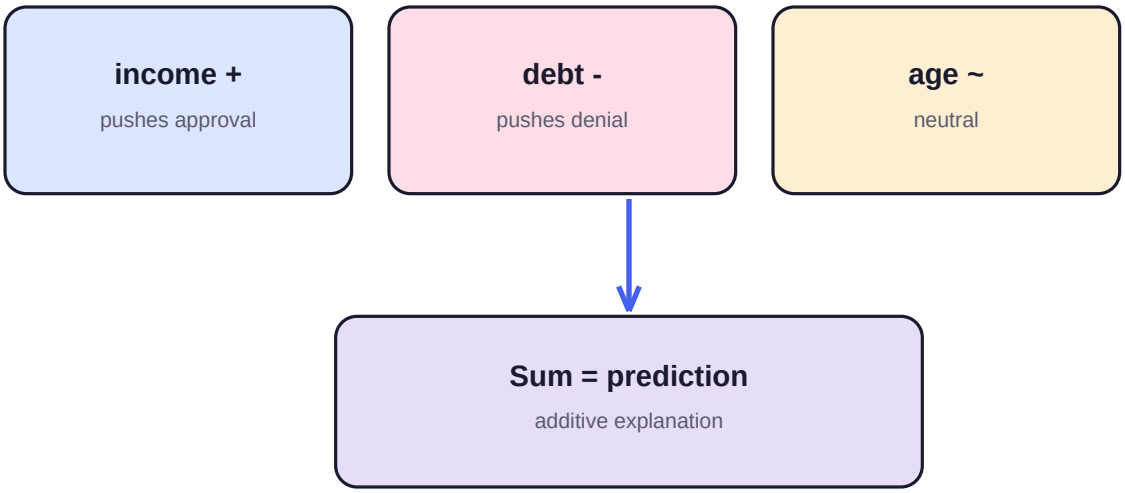
Local — 'why THIS prediction for THIS input?' — per-decision reasons.

Global explanations audit the model; local explanations justify individual decisions. You usually need both.

STEP 2/5

SHAP Values

Fair credit assignment from game theory.



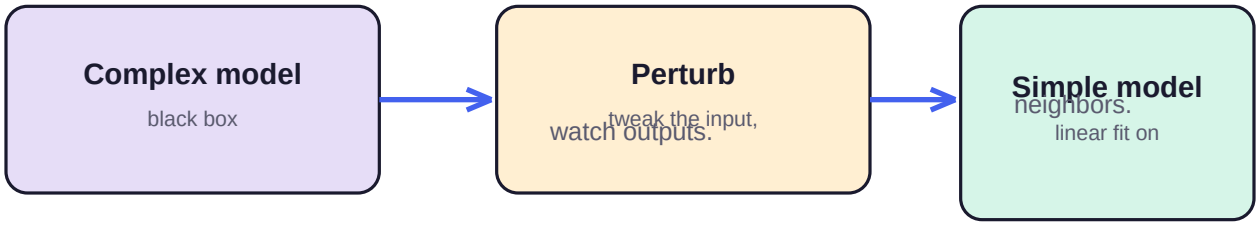
Each feature gets a Shapley value: its marginal contribution averaged over all feature orderings.

SHAP is consistent and theoretically grounded — but expensive on large models; use sampling or tree-specific fast paths.

STEP 3/5

LIME

Explain locally with a simple surrogate.



LIME fits an interpretable model on perturbed samples near the input. Fast and model-agnostic — but sampling noise makes explanations jittery.

STEP 4/5

Saliency and Attention Maps

Show WHERE the model looked.

Saliency

gradient of output w.r.t. input pixels — highlights influential regions.

Attention

weight heatmaps in transformers — which tokens the model used.

Great for debugging ('the model looked at the watermark, not the tumor'). Smooth with Grad-CAM or integrated gradients for stability.

STEP 5/5

When Explanations Mislead

Explanations are models too — they can be wrong.

- Attention is not explanation: high weight does not prove the token caused the output.
- Surrogates approximate: LIME/SHAP explain the neighborhood, not the true decision boundary.
- Explanations can be gamed: adversarial models can hide bias while producing innocent-looking SHAP plots.
- Confirmation bias: humans accept explanations that match their priors and ignore the rest.
- Use explanations to generate hypotheses, then verify with ablations and counterfactuals.

STEP ✓I✓

Key Takeaways

Explainable AI — Remember

- Global explanations audit the model; local ones justify single decisions.
- SHAP gives principled additive credit per feature; LIME fits local surrogates fast.
- Saliency/attention maps show where the model looked — great for debugging.
- Explanations can mislead: attention isn't proof, surrogates approximate, plots can be gamed.
- Treat explanations as hypotheses; verify with ablations and counterfactuals.

Keep exploring

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev